

CDAC Mumbai

Subject: AI Compute Platform

Date : 05-08-2026

Maximum Marks: 40

Title : Smart Water Distribution and Resource Analytics System

Problem Statement

A metropolitan water authority collects large volumes of data from **water treatment plants, distribution stations, smart meters, pipeline monitoring systems, and consumer usage records**. The authority plans to develop a **Smart Water Distribution Analytics Platform** using **Apache Spark** to optimize water supply, detect leakages, monitor water quality, analyze consumption patterns, and predict future resource demand.

As a **Big Data AI Engineer**, design, implement, analyze, and deploy the complete solution using **Apache Spark and PySpark**.

Dataset Details:

Dataset Name: Smart_Water_Distribution_Analytics.csv

Column Name	Description
Record_ID	Unique Record Identifier
Supply_Date	Water Supply Date
Zone	Distribution Zone
Distribution_Station	Distribution Station
Water_Source	Source of Water
Supply_Volume_Liters	Total Water Supplied
Consumption_Liters	Total Water Consumed
Water_Loss_Liters	Water Lost During Distribution
Pressure_PSI	Pipeline Pressure
Pipeline_ID	Pipeline Identifier
Leakage_Events	Number of Leakage Events
Water_Quality	Water Quality Category
PH_Value	Water pH Value
Turbidity_NTU	Turbidity Measurement
Consumer_Count	Number of Consumers
Maintenance_Cost	Maintenance Cost
System_Status	Operational Status

Q1. Spark Environment Setup and Data Loading (4 Marks)

Develop a PySpark program to perform the following tasks:

a) Initialize **SparkSession** and **SparkContext**.

b) Load the dataset **Smart_Water_Distribution_Analytics.csv**.

c) Display

- Dataset Schema
- First 10 Records
- Total Number of Records

d) Display descriptive statistics of numerical attributes.

Q2. RDD Programming and Transformations (6 Marks)

Create RDDs from the dataset and perform the following operations.

a) Create an RDD from the dataset.

b) Perform at least **three RDD Transformations** such as

- map()
- filter()
- flatMap()

c) Perform at least **three RDD Actions** such as

- count()
- collect()
- reduce()

d) Calculate **total water supplied** for each zone using Key-Value Pair RDD.

Q3. Spark DataFrame Operations (8 Marks)

Perform the following DataFrame operations.

a) Select the columns

- Zone
- Distribution_Station
- Supply_Volume_Liters
- Consumption_Liters

b) Filter records where

- Water Loss > 1000 Liters
- Leakage Events > 2

c) Sort records based on Water Loss in descending order.

d) Group records by Zone.

e) Calculate

- Total Water Supplied
- Total Water Consumed
- Total Water Loss
- Average Pipeline Pressure
- Average Maintenance Cost

Q4. Exploratory Data Analysis using Spark SQL (10 Marks)

Create a Temporary View and execute Spark SQL queries to answer the following.

- Identify the **Top 5 Zones** with maximum water consumption.
- Calculate **Zone-wise Water Loss Statistics**.
- Determine the **Distribution Station Efficiency**.
- Generate **Monthly Water Supply Trends**.
- Generate **Monthly Water Consumption Trends**.
- Find the zones with maximum leakage events.
- Display average pH value and turbidity for each water source.

Q5. ETL Pipeline Development (4 Marks)

Design and implement an ETL Pipeline.

The pipeline should

- Extract data from CSV.
- Remove duplicate records.
- Handle missing values.
- Convert data into appropriate data types.
- Save the processed dataset into **Parquet** format.

Draw the ETL workflow.

Q6. Machine Learning using Spark MLlib (4 Marks)

Develop a Machine Learning model using Spark MLlib to perform **ONE** of the following.

Option A

Predict future water consumption demand.

Option B

Predict pipeline leakage risk.

The implementation should include

- Feature Selection
- Model Training
- Prediction
- Performance Evaluation

Mention the evaluation metric used.

Q7. Deployment using DevOps Tools (4 Marks)

Perform the following deployment tasks.

- a) Create a GitHub repository and upload the complete project.
- b) Explain the CI/CD pipeline using GitHub Actions/Jenkins/GitLab CI.